ANALYSIS REPORT

RCIM Model-Bank Reproduction Overnight Gap-Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the overnight `RCIM Model-Bank Reproduction` campaign prepared in:

- `doc/reports/campaign_plans/track_1/harmonic_wise/2026-04-13-00-55-21_track1_overnight_gap_closure_campaign_plan_report.md`
  The package executed `20` repository-owned harmonic-wise validation runs through one coordinated launcher:
- completed runs: `20`
- failed runs: `0`
- execution window: `2026-04-13T01:42:34+02:00` to `2026-04-13T02:00:49+02:00`
- campaign artifact root: `output/training_campaigns/track1/harmonic_wise/forward/track1_overnight_gap_closure_campaign_2026_04_13_01_02_23/` The overnight batch was organized into four logical experiment blocks:
- `Campaign A` : low-order `HGBM` ladder
- `Campaign B` : late-harmonic repair ladder
- `Campaign C` : `RandomForest` counterfactuals
- `Campaign D` : engineered-term recovery

Objective And Outcome

The campaign had four concrete questions

1. can the current harmonic-wise baseline improve through stronger low-order treatment on `h0` and `h1` ;
2. can late-harmonic specialization reduce the remaining paper-table pressure around `162` and `240` ;
3. does `RandomForest` become competitive when tuned around the open harmonics;
4. do engineered operating-condition terms help when combined with targeted harmonic overrides instead of broad promotion. Outcome:

- the batch completed successfully with `20/20` runs and no observed launcher failures;
- the best test mean percentage error improved from the previous repository best `8.877%` to `8.774%` ;
- the improvement is real but still small, so the support branch for `Target A` remains `not_yet_met` against the paper threshold `4.7%` ;
- the strongest alternatives are still `HGBM` , especially the low-order and late-harmonic repair ladders;

- `RandomForest` remained clearly non-competitive for the shared offline evaluator in this campaign;
- the engineered-term re-check did not justify promotion over the best no-engineering `HGBM` runs. Primary `RCIM Model-Bank Reproduction` interpretation:
- this campaign does not close `RCIM Model-Bank Reproduction` by winner selection alone;
- its value is to provide support evidence for the still-open exact-paper cells;
- the canonical closure status must still be read from the exact-paper Tables `3-6` report, not from the campaign-local ranking table.

Ranking Policy

The campaign winner uses the explicit ranking rule approved in the campaign plan:

- eligible winner set: all completed runs that emit the shared offline evaluator;
- primary metric: `test_mean_percentage_error_pct` ;
- first tie breaker: `test_curve_mae_deg` ;
- second tie breaker: `test_curve_rmse_deg` ;
- third tie breaker: lexical `run_name` ;
- direction: minimize.

  This policy is serialized under:
- `output/training_campaigns/track1/harmonic_wise/forward/track1_overnight_gap_closure_campaign_2026_04_13_01_02_23/campaign_leaderboard.yaml`

Ranked Completed Runs

Rank	Config	Block	Test MPE [%]	Curve MAE [deg]	Curve RMSE [deg]
1	`track1_hgbm_h01_shallow_regularized`	A	8.774	0.002578	0.002776
2	`track1_hgbm_h162_h240_repair`	B	8.795	0.002578	0.002776
3	`track1_hgbm_h013_deeper_low_order`	A	8.954	0.002626	0.002823
4	`track1_hgbm_h081_h162_h240_repair`	B	8.959	0.002600	0.002797
5	`track1_hgbm_h01_ultradeep_guarded`	A	8.996	0.002672	0.002869
6	`track1_hgbm_h013_h162_h240_joint`	B	9.067	0.002674	0.002870
7	`track1_hgbm_h156_h162_h240_repair`	B	9.162	0.002665	0.002866
8	`track1_hgbm_h039_h162_h240_bridge`	B	9.167	0.002665	0.002866
9	`track1_hgbm_h01_deeper_low_order`	A	9.199	0.002697	0.002890
10	`track1_hgbm_h240_extreme_focus`	B	9.273	0.002703	0.002903
11	`track1_hgbm_h0139_low_order_anchor`	A	9.288	0.002738	0.002932
12	`track1_hgbm_h014078_low_order_anchor`	A	9.305	0.002744	0.002939
13	`track1_hgbm_h162_h240_engineered_recheck`	D	9.408	0.002655	0.002857
14	`track1_hgbm_h013_engineered_recheck`	D	9.501	0.002650	0.002852
15	`track1_hgbm_h01_engineered_recheck`	D	9.532	0.002671	0.002873
16	`track1_rf_h01_h081_engineered_recheck`	D	10.936	0.003234	0.003426
17	`track1_rf_h081_focus`	C	11.166	0.003315	0.003500
18	`track1_rf_h156_h162_h240_focus`	C	11.176	0.003318	0.003502
19	`track1_rf_full_rcim_reference`	C	11.183	0.003318	0.003502
20	`track1_rf_h01_focus`	C	11.190	0.003319	0.003503

Config	Validation MPE [%]	Oracle Test MPE [%]	Delta Vs `8.877%` Baseline
`track1_hgbm_h01_shallow_regularized`	9.611	2.749	-0.103
`track1_hgbm_h162_h240_repair`	9.448	2.749	-0.082
`track1_hgbm_h013_deeper_low_order`	9.479	2.749	+0.078
`track1_hgbm_h081_h162_h240_repair`	9.361	2.749	+0.082
`track1_hgbm_h01_ultradeep_guarded`	9.804	2.749	+0.119

Best Run Per Block

Block	Best Config	Test MPE [%]	Interpretation
A	`track1_hgbm_h01_shallow_regularized`	8.774	Mildly regularized low-order specialization worked better than deeper aggressive variants.
B	`track1_hgbm_h162_h240_repair`	8.795	Late-harmonic repair is still useful and nearly tied with the winner.
C	`track1_rf_h081_focus`	11.166	`RandomForest` remained far from competitive on the shared evaluator.
D	`track1_hgbm_h162_h240_engineered_recheck`	9.408	Engineered terms did not recover enough value to beat the best no-engineering runs.

Support-Branch Winner

The explicit campaign winner is

- `track1_hgbm_h01_shallow_regularized`

  Its result was:

- test mean percentage error: `8.774%`
- test curve MAE: `0.002578 deg`
- test curve RMSE: `0.002776 deg`
- oracle test mean percentage error: `2.749%`
- previous shared-evaluator best: `8.877%`
- absolute improvement versus previous best: `0.103 percentage points`
- Target A threshold: `4.7%`
- Target A status: `not_yet_met`

  This run is useful as support evidence because it is the lowest-error completed candidate under the shared offline evaluator, but it is not the canonical `RCIM Model-Bank Reproduction` closure signal.

Primary RCIM Model-Bank Reproduction Readout

For `RCIM Model-Bank Reproduction`, the important question after this campaign is not only which run won locally, but which paper-table cells can be treated as better supported for the next exact-paper repair pass.

The main campaign conclusions for that purpose are:

- `h0 / h1` remain the strongest support direction for the open low-order

  cells at `0` and `1`;

- `h162 / h240` remain the strongest isolated late-harmonic repair direction;
- `RandomForest` did not earn additional priority under the current

  coefficient-based support branch;

- engineered features did not justify promotion into the next `RCIM Model-Bank Reproduction`

  repair cycle.

Interpretation By Experiment Block

1. Low-Order Treatment Still Matters Most

The campaign winner came from the low-order `HGBM` ladder.

Important evidence

- `track1_hgbm_h01_shallow_regularized`: `8.774%`
- `track1_hgbm_h013_deeper_low_order`: `8.954%`
- `track1_hgbm_h01_ultradeep_guarded`: `8.996%`
- `track1_hgbm_h01_deeper_low_order`: `9.199%`

  Interpretation:

- `h0 / h1` are still the main leverage point for TE-level improvement;
- more capacity was not automatically better;
- the winning direction was not the deepest ladder, but the more controlled

  low-order regularized variant.

2. Late-Harmonic Repair Is Real But Secondary

The best late-harmonic run finished almost tied with the winner:

- `track1_hgbm_h162_h240_repair` : `8.795%`

  Interpretation:

- the late-harmonic block around `162` and `240` still matters;
- however, late-harmonic repair alone did not surpass the best low-order run;
- the next iteration should probably keep both directions alive instead of

  replacing one with the other.

3. `RandomForest` Did Not Reproduce The Paper Direction Here

All `RandomForest` runs ranked at the bottom:

- best `RF` : `track1_rf_h081_focus` at `11.166%`
- worst `RF` : `track1_rf_h01_focus` at `11.190%`

  Interpretation:

- under the current coefficient-based shared evaluator, `RF` is not a viable

  promotion candidate;

- this does not invalidate the paper's per-harmonic family choices, but it

  does show that the current repository harmonic-wise path still strongly favors `HGBM` over `RF` .

4. Engineered Terms Did Not Earn Promotion

The best engineered run was

- `track1_hgbm_h162_h240_engineered_recheck` : `9.408%`

  Interpretation:

- the engineered operating-condition terms remain weaker than the best

  no-engineering `HGBM` configurations for this branch;

- the overnight batch therefore does not justify making engineered terms the

  new `RCIM Model-Bank Reproduction` default.

Recommended Next Step

The batch improved the supporting harmonic-wise evaluator, but it did not change the canonical `RCIM Model-Bank Reproduction` closure verdict by itself.

The most defensible next step is

1. keep `track1_hgbm_h01_shallow_regularized` only as the current best

   support-branch reference;

2. keep `track1_hgbm_h162_h240_repair` as the strongest late-harmonic repair

   companion;

3. feed both directions into the next exact-paper open-cell repair plan;
4. if those repair-driven iterations still plateau, move the next research

   step to a new target-parameterization implementation rather than another winner-centric `RF` or engineered-feature retry.

Artifact References

- `output/training_campaigns/track1/harmonic_wise/forward/track1_overnight_gap_closure_campaign_2026_04_13_01_02_23/campaign_leaderboard.yaml`
- `output/training_campaigns/track1/harmonic_wise/forward/track1_overnight_gap_closure_campaign_2026_04_13_01_02_23/campaign_best_run.yaml`
- `output/training_campaigns/track1/harmonic_wise/forward/track1_overnight_gap_closure_campaign_2026_04_13_01_02_23/campaign_best_run.md`